HDOC Day-6 Activity

Question 1:

A square handkerchief has side s centimeters. Write a C program to find the length of lace required around it.

Step 1: Understanding the Problem

- **Input:** Side s of the square handkerchief in centimeters.
- **Output:** Length of lace required around the handkerchief in centimeters.
- **Formula:** Lace required = Perimeter of square = 4 x s

Step 2: Standard Test Case Table

Test Case	Input	Expected Output	Actual Output	Result
1	s = 5 cm	20.00 cm	20.00 cm	Pass
2	s = 12.5 cm	50.00 cm	50.00 cm	Pass
3	s = 20 cm	80.00 cm	80.00 cm	Pass

Step 3: Algorithm and Evaluation

1. Start.

2. Declare side and lace as float variables.

3. Read side s from the user.

4. Calculate lace = 4 * side.

5. Display the required length of lace.

6. Stop.

Evaluation: The algorithm directly uses the perimeter formula of a square. For each valid side value, it produces the required lace length correctly.

Step 4: C Programming

```c
#include <stdio.h>

int main()
{
    float side, lace;

    printf("Enter side of square handkerchief (cm): ");
    scanf("%f", &side);

    lace = 4 * side;

    printf("Length of lace required = %.2f cm\n", lace);

    return 0;
}
```

Step 5: Compilation and Execution

The program was compiled and executed in Kali Linux and evaluated against the test cases above.

Step 6: Kali Linux Compilation and Execution

```
(kali㉿kali)-[~]
$ nano handkerchief.c
(kali㉿kali)-[~]
$ gcc handkerchief.c -o handkerchief
(kali㉿kali)-[~]
$ ./handkerchief
Enter side of square handkerchief (cm): 5
Length of lace required = 20.00 cm
```

Test Case 2:

```
(kali㉿kali)-[~]
$ ./handkerchief
Enter side of square handkerchief (cm): 12.5
Length of lace required = 50.00 cm
```

Test Case 3:

```
(kali㉿kali)-[~]
$ ./handkerchief
Enter side of square handkerchief (cm): 20
Length of lace required = 80.00 cm
```

Question 2:

Write a C program to find the base, area, and perimeter of a parallelogram when its height, adjacent side, and area are given.

Step 1: Understanding the Problem

- **Input:** Height h, adjacent side a, and area A of the parallelogram.
- **Output:** Base b, area A, and perimeter P.
- **Formula:** Base b = A / h; Perimeter P = 2 x (b + a). The area is already given as input.

Step 2: Standard Test Case Table

Test Case	Input	Expected Output	Actual Output	Result
1	h=5, a=4, A=30	b=6.00, A=30.00, P=20.00	b=6.00, A=30.00, P=20.00	Pass
2	h=8, a=7, A=80	b=10.00, A=80.00, P=34.00	b=10.00, A=80.00, P=34.00	Pass
3	h=2.5, a=6, A=25	b=10.00, A=25.00, P=32.00	b=10.00, A=25.00, P=32.00	Pass

Step 3: Algorithm and Evaluation

1. Start.

2. Declare height, adjacent_side, area, base and perimeter as float variables.

3. Read height, adjacent side and area.

4. Calculate base = area / height.

5. Calculate perimeter = 2 * (base + adjacent_side).

6. Display base, area and perimeter.

7. Stop.

Evaluation: Since area and height are known, the base is obtained from A = b x h. The perimeter is then calculated using P = 2(b + a). The algorithm gives the expected result for all prepared test cases.

Step 4: C Programming

```c
#include <stdio.h>

int main()
{
    float height, adjacent_side, area, base, perimeter;

    printf("Enter height: ");
    scanf("%f", &height);

    printf("Enter adjacent side: ");
    scanf("%f", &adjacent_side);

    printf("Enter area: ");
    scanf("%f", &area);

    base = area / height;
    perimeter = 2 * (base + adjacent_side);

    printf("Base = %.2f\n", base);
    printf("Area = %.2f\n", area);
    printf("Perimeter = %.2f\n", perimeter);

    return 0;
}
```

Step 5: Compilation and Execution

The program was compiled and executed in Kali Linux and evaluated against all three test cases.

Step 6: Kali Linux Compilation and Execution

```
(kali㉿kali)-[~]
$ nano parallelogram.c
(kali㉿kali)-[~]
$ gcc parallelogram.c -o parallelogram
(kali㉿kali)-[~]
$ ./parallelogram
Enter height: 5
Enter adjacent side: 4
Enter area: 30
Base = 6.00
Area = 30.00
Perimeter = 20.00
```

Test Case 2:

```
(kali㉿kali)-[~]
$ ./parallelogram
Enter height: 8
Enter adjacent side: 7
Enter area: 80
Base = 10.00
Area = 80.00
Perimeter = 34.00
```

Test Case 3:

```
(kali㉿kali)-[~]
$ ./parallelogram
Enter height: 2.5
Enter adjacent side: 6
Enter area: 25
Base = 10.00
Area = 25.00
Perimeter = 32.00
```